

Linear Programming Course Summary

Generated by Scribe.AI

December 4, 2024

1 Key Terms

1.1 linear programming fundamentals

- **Slack Variable:** *A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. It represents the difference between the left-hand side and right-hand side of the inequality.*

1.2 solution space analysis

- **Feasible Region:** *The set of all points that satisfy all the constraints of a linear programming problem.*
- **Vertex:** *A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In higher dimensions, it is the intersection of multiple constraint hyperplanes.*

1.3 simplex method algorithms

- **Basic Variable:** *In the Simplex method, a variable that is expressed explicitly in terms of non-basic variables in a dictionary.*
- **Non-basic Variable:** *In the Simplex method, a variable that is set to zero in a given iteration of the algorithm.*
- **Pivot Operation:** *The process of exchanging a basic and a non-basic variable in the Simplex method, updating the dictionary to move to a new vertex.*
- **Dictionary:** *A representation of a linear programming problem where basic variables are expressed as functions of non-basic variables. It is updated during the Simplex method iterations.*
- **Auxiliary Problem:** *A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem.*

2 Problem Types

2.1 linear programming fundamentals

- **Finding an Initial Feasible Solution:** *The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.*
- **Using Dictionaries to Represent Solutions:** *The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.*

2.2 advanced linear programming

- **Finding an Initial Feasible Solution:** *The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.*
- **Handling Infeasible Origins:** *The origin $(0,0)$ may not satisfy all constraints in a linear programming problem. Techniques like introducing an auxiliary problem are used to find a feasible starting point.*
- **Optimizing in Higher Dimensions:** *The Simplex method is extended to solve linear programming problems with more than two variables, requiring iterative pivoting operations to move between vertices of the feasible region in higher-dimensional space.*
- **Using Dictionaries to Represent Solutions:** *The Simplex method uses dictionaries to represent the current solution and to perform pivot operations efficiently. Each dictionary corresponds to a vertex of the feasible region.*

3 Algorithm Solutions

3.1 linear optimization techniques

- **Simplex Method:** *Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This corresponds to moving from vertex to vertex in the feasible region.*
- **Auxiliary Problem:** *When the origin is not feasible, introduce a new variable x_0 and modify the constraints to create an auxiliary problem that maximizes $-x_0$. Solve this auxiliary problem using the Simplex method. If the optimal solution has $x_0 = 0$, then the solution is feasible for the original problem.*